

Contributions of GPS Permanent Station and Seismological Observatory to Geophysical Studies at Maitri

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ABSTRACT

On going research activities at Antarctica were successfully carried out during XXIII IAE for the long-term scientific objectives of NGRI. The data from GPS station continues to contribute to Scientific Committee on Antarctic Research (SCAR) and International Earth Rotation Service (IERS). The seismological observatory continues to contribute to National Earthquake Information Center (NEIC), International Seismological Center (ISC) plays a pivotal role in the Global Seismographic Network in Antarctica. Global positioning system (GPS) measurements at Maitri, Indian Antarctic station, have been carried out for the period 1997 to till date. The International GNSS Service (IGS) stations, in different plates, such as Casey, Davis, Kerguelen, Seychelles, Coco, Hartebeesthoek, Yagadee, and Tidbinbilla were chosen for global network solution using Bernese 4.2 software. The GPS data from 1997 to 2005 of all these IGS stations in and around Antarctica along with Maitri were used to estimate the time series, baseline lengths and their changes and velocity vectors of individual sites. The results have enabled to obtain a deeper insight into the driving mechanism and the causative factors for Indian plate kinematics. During this expedition i.e., from January 2004 to February 2005 continuous seismic data was also acquired for processing and analysis. The final data analysis of this data has been completed using SEISAN software and reported to ISC, United Kingdom as per their time schedule up to December 2004. The number of earthquakes recorded as global events and near-events in and around Antarctica during the period of expedition is discussed in this report.

INTRODUCTION

The India–Eurasia collision zone has been the focus of intense studies by the Space Geodetic Community for Indian Plate Motion. So far space geodetic studies by GPS have been made in southern and central India that bears on the rigidity of the Indian plate, whose velocity data for two sites fit

the rigid plate model within uncertainties, estimated at about 3.7 cm/yr. for Hyderabad and 3.9 cm/yr. for Bangalore respectively (Malaimani et al, 2000). Since Space geodesy can test many otherwise untestable predictions and boundary conditions, GPS-Geodesy is extended to envelop the large Indian Ocean and Antarctica (Stein et al, 2002). Despite the extensive studies to date, major issues remain unresolved. Motions across some of the plate boundaries seem well constrained, as implied by good agreement between space geodetic and geologic models. In others, apparent discrepancies exist. Very few studies have been conducted on the larger oceanic part of the Indian plate using space geodesy (Malaimani et al, 2005; Ravikumar et al 2005). To holistically determine the kinematics of the Indian Ocean Basin between Antarctica and India, the data available are very sparse and characterization and the delineation of the plate boundaries especially in the Indian Ocean are poor. This study addresses several of these issues mentioned as follows.

(1) How rigid is the Indian Plate and Indian Ocean Basin? (2) Does relatively high level of intraplate seismicity on the oceanic part of the Indian Plate indicates internal deformation in excess of other plates? and, (3) Is this related to Indo-Eurasian collision and the uplift of Himalayas?

Therefore, GPS geodesy is used to improve the understanding of the complex plate motions, diffuse and poorly located plate boundaries, and striking intraplate deformation that characterize the Indian Ocean basin.

WORK CARRIED OUT DURING THE EXPEDITION

Operated and acquired the GPS and Seismic data with the GPS permanent station and Seismological Observatory comprising of Ashtech, Trimble and Benchmark ACT GPS receiver and Reftek Digital broadband system and analog portacorder. Due to intermittent failure of Ashtech GPS receiver, frequent repairs were necessary and hence as a stand by Trimble GPS receiver was installed. In this study, we used data acquired by Ashtech GPS receiver up to February 2004 and data acquired by Trimble GPS receiver from March 2004 till December 2004 as used in this study. The receiver worked continuously and data was acquired. The daily data is of the size of 3.2 Mega bytes (approximately), with 15 sec sample rate for 9 channels. Trimble GPS receiver was replaced with the ACT Benchmark 12 channel GPS receiver with IGS choke ring GPS antenna. For data downloading Pentium III computer was installed and its RS-232 port was used for downloading the data via "HYPER" terminal software. The receiver acquired the data continuously was converted into RINEX files. The acquisition and pre-processing of data are done regularly with maximum care. The data was processed using Bernese 4.2 version at NGRI.

To improve the system performance we have opened the seismic vault and completed the sensor leveling and mass centering by locking and unlocking. The entire seismic system was refurbished for satisfactory operation. After doing this, we observed that the daily file size improved from 20 to 25 MB (as it was earlier). There was improvement in the signal level also.

During July 2004 the portacorder failed and the portacorder drum was not rotating. The drum motor was changed and the relevant IC has been changed in the main RV320b board and again the portacorder was put in service. Temperature was maintained properly in the observatory and seismic vault. The temperature was maintained from 20° to 25° C with the help of the temperature sensor and thermometer. A few studies (Navin Chander et al., 2002; Navinchader et al., 2003; Malaimani et al., 1998; Maiaimani et al., 1999; Murthy et al., 2001; Premkishore et al., 2002) were carried out on GPS & seismology, who discussed the seismotectonics and geodynamical processes between India and Antarctica using GPS and three component seismograms recorded at Maitri.

PERMANENT GPS STATION AT MAITRI

During 23rd IAE i.e., from January 2004 to February 2005 continuous GPS data was acquired for processing and analysis. During the period of expedition, Ashtech 12-channel GPS receiver was in operation upto February 2004, while Trimble 9-channel dual frequency geodetic receivers was in operation in the same location from March 2004 with 15 sec sampling interval. Global positioning system (GPS) measurements at Maitri, Indian Antarctic station, have been carried out for the period from 1997 till date to study the temporal evolution of baseline length between Hyderabad and Maitri, and to understand the tectonic and geodynamical processes taking place in Indian Ocean.

GPS Data Acquisition and Analysis

The Global network of GPS stations include Maitri (MAIT), Casey (CAS1) and Davis (DAV1) in the Antarctic plate, Seychelles (SEY1) in Somalian plate, Cocos (COCO) on edge of Indo-Australia diffuse plate boundary, Yarangadee (YAR1), Tidbinbilla (TID2) and Australian plate and Hartebeesthoek (HRAO) in African plate. The GPS data brought back to NGRI and processed and analyzed using Bernese version 4.2. to estimate the time series, baseline lengths and their changes and velocity vectors of individual sites. The GPS data was contributed to SCAR. Figure (1) shows choke ring GPS antenna at Maitri.

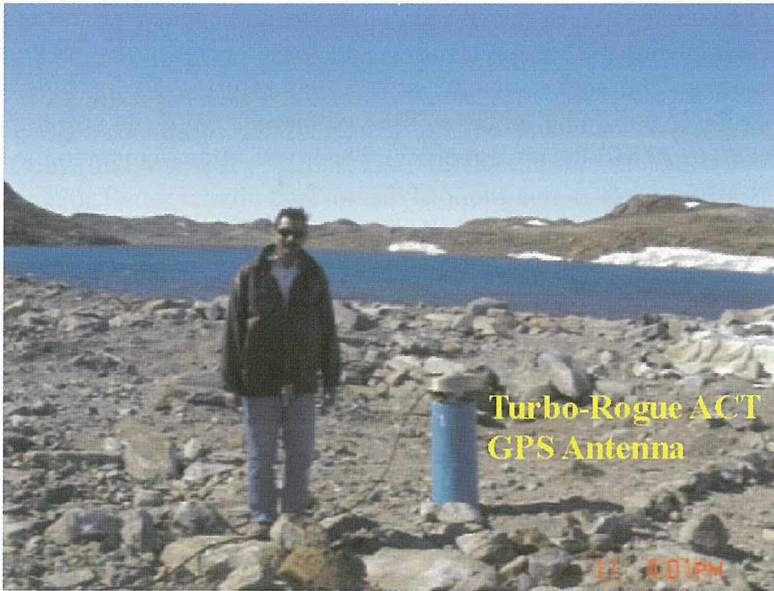


Fig. 1: Choke ring GPS antenna at Maitri

GPS Data Processing

Bernese version 4.2 scientific software is used for data processing. Data processing carried out in several steps include conversion of data, orbital part, code data processing, phase data processing and finally estimated the precise baseline lengths and site coordinates. The input observables are the L1 carrier, L2 carrier frequency, C/A code from the RINEX observation files and the precise orbit files from the IGS data center. After processing the data using Bernese 4.2 software, we will get the output file named GPSEST, which contains estimated baseline length and coordinates for further analysis for strain accumulation and velocity vectors (Hugentobler, U., et al, 2001).

THE SEISMOLOGICAL OBSERVATORY AT MAITRI, ANTARCTICA

Focused study on the Indian Ocean and South Sandwich Islands Region Fig. (3) becomes very important in the present scenario of interplate and intraplate earthquakes that do occur frequently in India and nearby regions, forcing us to re-evaluate the rigidity of the Indian plate as well as Indian Ocean Margin. GPS-Geodesy and Seismic Studies are extended to envelop

the large Indian Ocean lithosphere. In order to monitor the seismic activity in and around Antarctica and Indian Ocean, a reconnaissance survey for site selection and feasibility of operation of seismological observatory in Antarctica was initiated during 16th Indian Antarctic Expedition and a Digital Short Period System was installed during 17th expedition. The station was fully commissioned on 26 January 1998 with analog and short period digital systems. This observatory was further upgraded with the installation of Broadband seismic system in 20th IAE, comprising of a Broadband Sensor CMG-3ESP (GURALP make with 50 Hz to 100 sec flat response) and Reftek Data Acquisition System.

The permanent seismic station at Maitri has gone global by figuring in Answer. (Antarctic Seismic Web Resource) as shown in Fig. (4), with the station code MAIT. After final analysis of seismic broad band digital data, the complete phase data had been reported to ISC, to be included in the global data center for global access by any other agency and can be accessed at: <http://www.isc.ac.uk>

The Seismic sensors (S-13 for short period & analog recorders and Guralp CMG-3ESP for Broadband System) are installed in the underground vault behind Tirumala Hut. The Data Acquisition System (DAS) – REFTEK High Resolution Digital Data Acquisition System RT 72A 121-03 from Refraction Technology Inc., USA interfaced with a P.C., Analog Recorder – vertical component analog RV-320B Portacorder and a GPS receiver – are all installed in the Tirumala hut. The Sensor pit and the lab are maintained at about 15 to 20 degrees temperature using electrical heaters. The data acquired was processed using SEISAN software (Havskov and Ottemoller et al., 2005). During this expedition i.e., from January 2004 to February 2005 continuous seismic data has been recorded and the same has been archived and stored in CDs for further processing and analysis.

Using SEISAN software the final analysis of this data was completed and reported to International Seismological Centre, United Kingdom up to December 2004. The data acquisition during wintering period, the Preliminary Determination of Epicenters (PDE) reports by NEIC are referred to while processing the data using SEISAN 7.2, PCSUDS and REFTEK utilities software. After completion of the analysis, the arrival times, origin times and the corresponding hypo central parameters of the processed events were used to compile the MAITRI Seismological Bulletin for the year 2004.

RESULTS AND DISCUSSION

From the GPS Results

Figure (2) shows the velocity and the direction of movement of stations. Estimated site velocities are given in Table-1. Data processing resulted in the estimation of all site coordinates and the baseline lengths from Kerguelen to all other stations. The estimation of very long baseline lengths and their changes between India and Antarctica give insight into the signatures of crustal deformation and strain accumulation in the south of Indian peninsula.

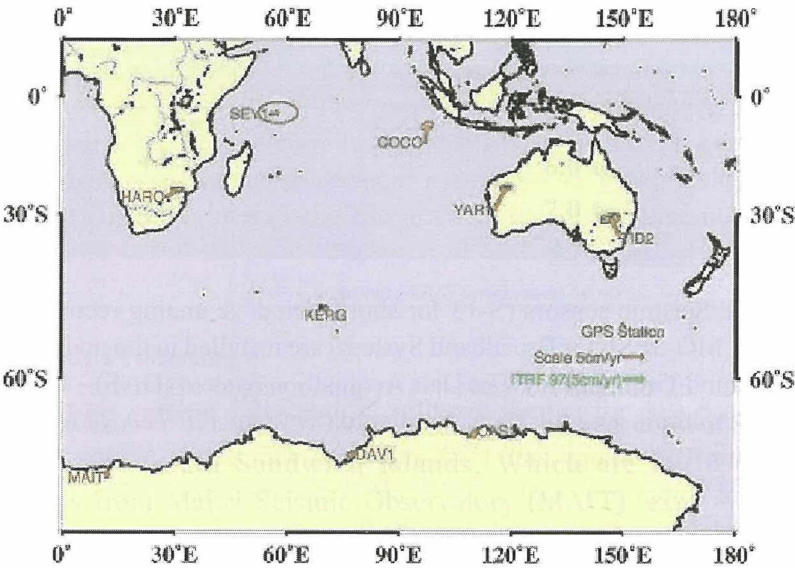


Fig. 2: GPS Velocity vectors of Global Network including Maitri GPS station

Table 1 : Estimated site velocities

Site	N-VEL(mm)	E-VEL(mm)	N-Error (mm)	E-Error (mm)
MAIT	11.30	2.30	1.009	1.768
CAS1	-12.00	7.60	2.222	4.019
DAV1	-2.90	-6.40	1.495	2.542
YAR1	50.80	27.90	5.59	2.49
TID2	43.50	-22.0	9.399	3.472
COCO	40.30	48.0	5.696	3.694
SEY1	3.90	17.80	22.794	11.718
HRAO	15.80	23.60	6.087	2.703

The overall analysis is that the estimated baseline lengths between Kerguelen and Maitri and other IGS stations namely Casey, Davis in Antarctica Plate are indeed shortening at the rates of 5.9 mm/yr, 4.4 mm/yr and 1.9 mm/yr respectively. The rate of change in the order of mm in the base line lengths between these stations indicate no significant change at the one standard deviation level and may be construed as the stations in the same plate are moving together. The increase in baseline length between Kerguelen (KERG) and Seychelles (SEY1) at the rate of 5.6 mm/yr clearly indicates that Seychelles (SEY1) is moving away from Kerguelen (KERG). The velocity vector of Seychelles (SEY1) shows the movement towards the Indian peninsula and this also agrees with the ITRF 97 estimation. If this rate of movement is verified upon, this may result in the increase in the strain accumulation in the southern Indian peninsula. GPS data at COCO suggests the high rate of movement that could be the result of excessive strain accumulation due to the Indo-Australian diffuse plate boundary forces acting upon this region.

From the Seismic Results

During the XXIII Indian Antarctic Expedition uninterrupted good quality Seismic analog & Digital Broad Band data were acquired. The data acquired was processed and analyzed at NGRI. For the Digital Broad Band seismic data, SEISAN software was used for the final processing and analysis. In the Indian Antarctic Station, Maitri, a permanent Seismological Station was commissioned in 1998 with an analog recorder. Subsequently, it has been upgraded with a Broad Band Digital System in January, 2001. The Broad Band Seismic data was contributed to ISC, U.K. We picked P and S phases in the seismograms and computed the epicentral distance from the difference of P&S arrival times. We then located the earthquakes using three component broadband seismic records using SEISAN software. The Processing of the data up to 2004 has been completed and the seismological Bulletin of Maitri Station has been published for 2004. The data processing for the year 2004 has been completed up to December 2004. The total number of earthquakes recorded from January to December 2004 was 270. Fig. 3 shows that the epicentral map of the events recorded during 2004.

Epimap of 270 earthquakes recorded at NGRI Seismic Observatory, Maitri, Antarctica during 23rd IAE during 2004

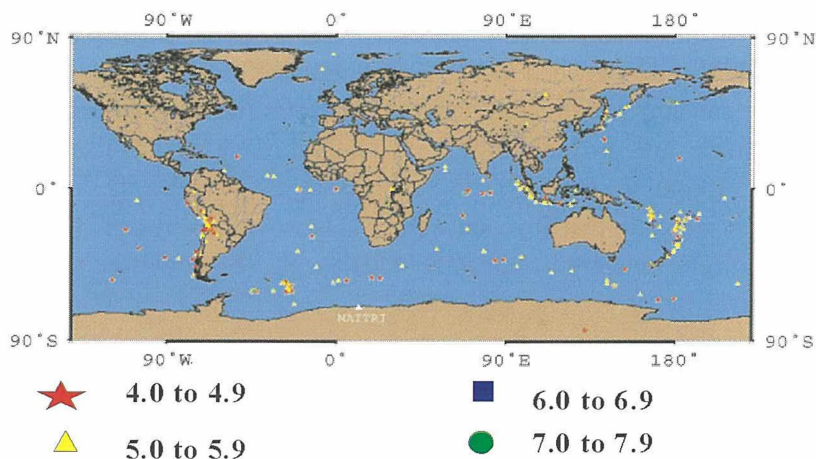


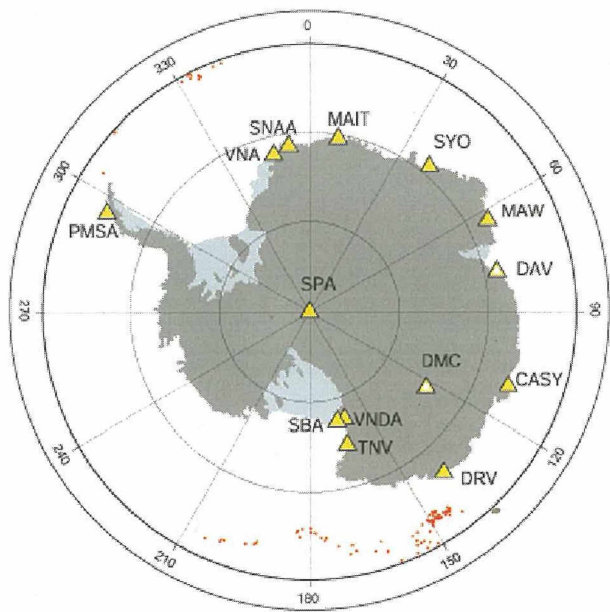
Fig. 3: Epicenter Map of Global events recorded at Maitri during the expedition

An earthquake of magnitude of 8.8 was recorded on 26th December 2004 at a distance of about $D=91.20^\circ$ located off the west coast of Sumatra. The area of South Sandwich Islands, Which are in 16° to 25° distances from Maitri Seismic Observatory (MAIT), experienced 28 earthquakes of Magnitudes ranging from 4.7 to 6.9. Scotia Sea region, which is in between 23° to 24° distances from Maitri, registered about 4 earthquakes ranging from 4.1 to 5.3 magnitudes. Another area of interest is the Indonesia / Sumatra Region which accounted for 19 earthquakes of Magnitude ranging from 5.0 to 8.8. Many earthquakes from south of Indian Ocean, South of South Africa, Japan, Fiji Islands, Chile-Argentina border region etc were also recorded. The area of South Sandwich Islands, Which are in 16° - 25° distance from Maitri Seismic Observatory (MAIT), experienced 28 earthquakes of Magnitudes ranging from 4.7 to 6.9. Monthly distribution of 270 earthquakes recorded at Maitri during 2007 is shown in Figure (5). During the period of expedition, the deadly Sumatra earthquake also has been recorded at Maitri and the waveform recorded is as shown in Figure 6.

Maitri in Antarctic Seismic Web Resource (AnSWeR)

Antarctic Broadband Seismic Observatories

Filled triangles=Existing stations, Open triangles=Proposed stations



CODE	NAME	INSTRUMENTATION	VAULT/ SITE	WHERE TO GET DATA
MAIT	Maitri Station	CMG-3ESP	Underground vault, bedrock	ISC

Fig. 4: Permanent Seismological Observatory in AnSWeR

Monthly distribution of 270 earth quakes recorded at Maitri in 2004

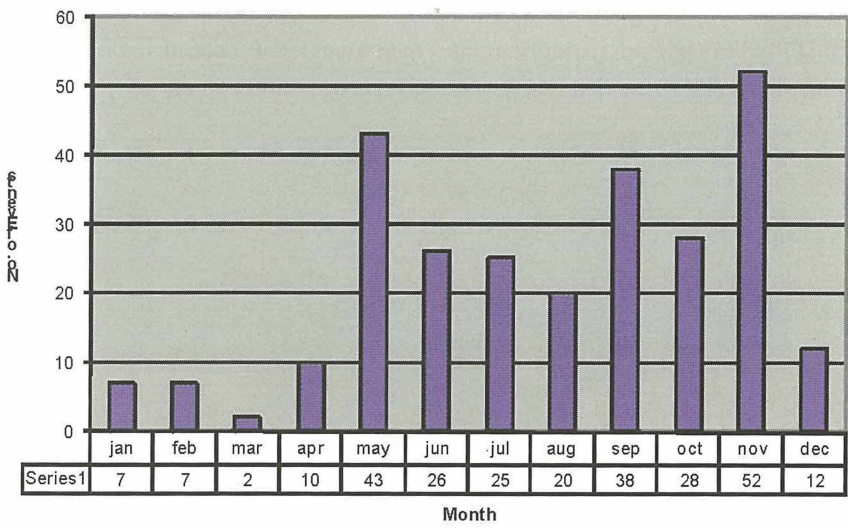


Fig. 5: Histogram showing the temporal distribution of global events

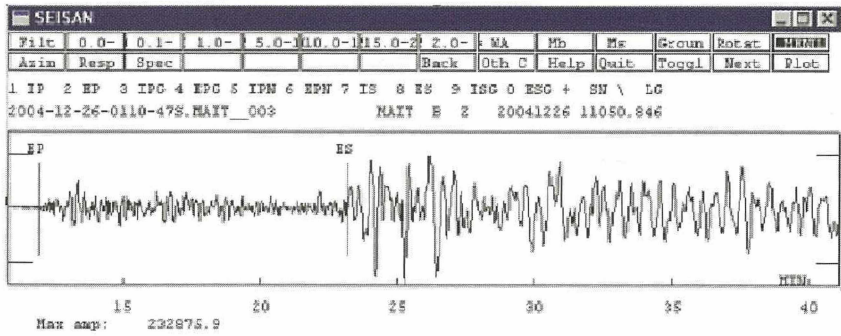


Fig. 6: Wave form of 26-12-2004 Sumatra Earthquake of magnitude $M_b=9.0$ recorded at Maitri Seismological observatory

CONCLUSIONS

This collocated GPS and Seismic studies enabled a deeper insight into the driving mechanisms responsible for Indian plate kinematics. Islands Seychelles and Coco are converging toward Indian plate The station

Yaragadee in Australian plate is moving away from Indian plate confirming the drifting of Indo-Australian plate, especially the western Australia and the emergence of a diffuse plate boundary between India and Australia. The GPS and the seismic result show the strain accumulation and deformation processes towards the Indian plate. The Stations Maitri, Casey, and Davis show a consistent movement within the Antarctic plate. The net result of these complexities is that the plate kinematics of the region remain less well described compared to other regions, and the issue of plate kinematics is inextricably linked to the question of plate rigidity.

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REFERENCES

1. Hugentobler, U., et al 2001: Bernese GPS Software Version 4.2 manual.
2. Havskov and Ottemoller et al., 2005: SEISAN The Earthquake Analysis Software, Ver 8.1.
3. Malaimani, E.C. et al., 2000: Indian plate kinematic by GPS-geodesy. *Earth Planets and Space*, 52 (10), 741-745.
4. Malaimani, E.C. et al, 2005: Present day Plate Kinematics between India and Antarctica through the Indian Ocean Basin by GPS-Geodesy – A holistic view, International Seminar on *Earth Science of East Antarctica (ESEA)*, 2005, New Delhi.
5. Ravikumar, N. et al, 2005: Evaluation of elastic strain accumulation in the Southern Indian peninsula between India and Antarctica by GPS-Geodesy, *Graticule 2005*, New Delhi.

6. Stein et al., 2002: The January 26, 2001 Bhuj earthquake and diffuse western boundary of the Indian plate, in plate boundary zones (ed), *Geodynamics series*, 30, AGU, Washington DC.
7. Navin Chander, et al., Indian Antarctic Expedition Seismological bulletin of Maitri station, Antarctica (2002).
8. G.B. Navin Chander, et al., Indian Antarctic Expedition Seismological bulletin of Maitri station, Antarctica (2003).
9. E.C Malaimani and L.Premkishore, Short-term GPS measurements at Maitri, Indian Antarctic station, Technical Publication No 14., 16th IAE(1998).
10. E.C.Malaimani, L.Premkishore and I.P.Raju., Establishment of permanent GPS tracking station At Maitri, Technical publication No 15., 17th IAE(1999).
11. Y.V.V.B.S.N. Murthy., Geophysical Activities at Maitri, Antarctica., Technical Publication No 17., 19th IAE(2001).
12. L. Prem Kishore, T.Chaitanya, A.Akilan, G.S.Srinivas, G.B.Navin Chander, S.V.R.R Rao, E.C.Malaimani and N.Ravikumar., Studies on Seismotectonics and Geodynamical Processes between Antarctica and India, Technical Publication No 18., 20th IAE(2002).